From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

# JEE (Main), 2014

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## CRT –III
## (Main)

| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | A | B | A |
| 2. | A | D | D |
| 3. | C | A | B |
| 4. | A | A | C |
| 5. | C | D | A |
| 6. | C | A | C |
| 7. | A | A | D |
| 8. | D | B | C |
| 9. | B | A | B |
| 10. | C | A | B |
| 11. | A | B | C |
| 12. | D | C | B |
| 13. | D | D | A |
| 14. | C | B | B |
| 15. | C | C | C |
| 16. | A | D | B |
| 17. | B | C | B |
| 18. | B | C | A |
| 19. | B | A | C |
| 20. | D | B | A |
| 21. | A | C | C |
| 22. | C | D | C |
| 23. | C | D | D |
| 24. | A | D | A |
| 25. | C | C | B |
| 26. | C | C | C |
| 27. | A | D | C |
| 28. | C | D | D |
| 29. | A | C | B |
| 30. | C | B | B |

# Physics

PART – I

1. $V = Es$ ;

$E = \frac{2sm}{qt^2}$

$\Rightarrow V = \frac{2s^2m}{qt^2}$ = 11.375 Volt

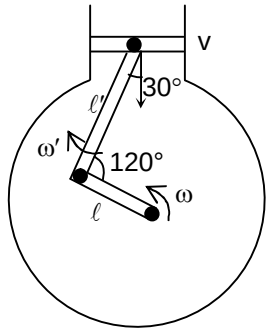
.2 $\frac{\ell}{\sin 30^\circ} = \frac{\ell'}{\sin 60^\circ}$

$\Rightarrow \ell' = \sqrt{3}\ell$

$\ell'\omega'\cos 30^\circ = v\cos 30^\circ$

$\ell\omega = \cos 60^\circ(v + \ell'\omega') = \cos 60^\circ(2v) = v$

$\omega = v/\ell = \frac{40\sqrt{3}}{0.1\sqrt{3}} = 400$ rad/s




3. for x<0

It will oscillate like SHM

Time period $T_1 = \pi\sqrt{\frac{m}{k}}$

For x>0

It will perform periodic motion under constant force

$T_2 = 2\sqrt{\frac{2E}{mg^2}}$

Hence time period $T = T_1 + T_2$

$T = \pi\sqrt{\frac{m}{k}} + 2\sqrt{\frac{2E}{mg^2}}$

4 $ev_0B = \frac{mv_0^2}{R}$

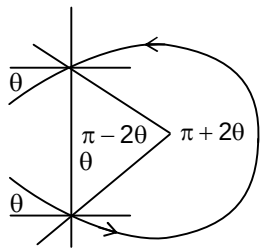
$B = \frac{mv_0}{eR}$

$\sqrt{a^2 + R^2} = b - R$

$R = \frac{b^2 - a^2}{2b}$

$B = \frac{2bmv_0}{(b^2 - a^2)e}$

5. Here the total time is given T
Time spent m the region is $(\pi + 2\theta)$
Hence total time spent
$= T\left(\frac{\pi + 2\theta}{2\pi}\right)$



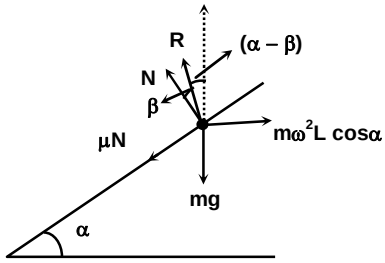

6. Torque of Internal force is always zero. All other quantities will vary.

8. Static friction depends on tendency.

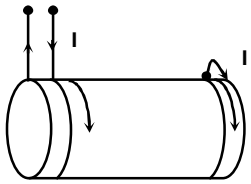
9. $\frac{ML^2}{12}\sin^2 60 \times 2 = \frac{3ML^2}{24}$

10. When impulse of external force is zero.

11. Balance forces.




12.




13. $y(x, t) = \left[e^{-\left(t-\frac{x}{b}\right)^2}\right]^a \Rightarrow v = b$

15. $2L = m \times (\lambda_1/2) = \frac{m}{2f}\sqrt{\frac{T}{9\mu}}$

$L = n \times (\lambda_2/2) = \frac{n}{2f}\sqrt{\frac{T}{4\mu}}$

$\Rightarrow \quad m = 3n = 3 \times 1 = 3$

16. $\cos\phi = 1 \Rightarrow \phi = 0° \Rightarrow \omega L = 1/\omega C$

17. $f_B = f - f_0 = \left(\frac{v + v_0 \sin 30°}{v - v_0 \sin 30°}\right) f_0 - f_0$

$= \frac{2v_0 f_0}{2v - v_0}$

18. $\frac{1}{2}mv^2 = h\nu$

19. $\beta = \lambda D/d$, $\lambda$ is minimum for violet.

20. $\vec{V}_{PG} = \vec{V}_{PC} + \vec{V}_{CG} = 2x\hat{j} + 2\hat{i}$

Hence, for the paritcle,
$V_x = 2m/s$ and $V_y = 2x = 4t$
$a_x = 0$ $a_y = 4m/s^2$
$x = 2t$, $y = 2t^2$
$\Rightarrow$ $a_{Total} = 4m/s^2$
(a) (along Y-axis)

and trajectory $\Rightarrow y = 2.\frac{x^2}{4} = \frac{x^2}{2}$

when $\left|\vec{a}_\tau\right| = \left|\vec{a}_N\right|$, a cos $\theta$ = a sin $\theta \Rightarrow \theta = 45^o$

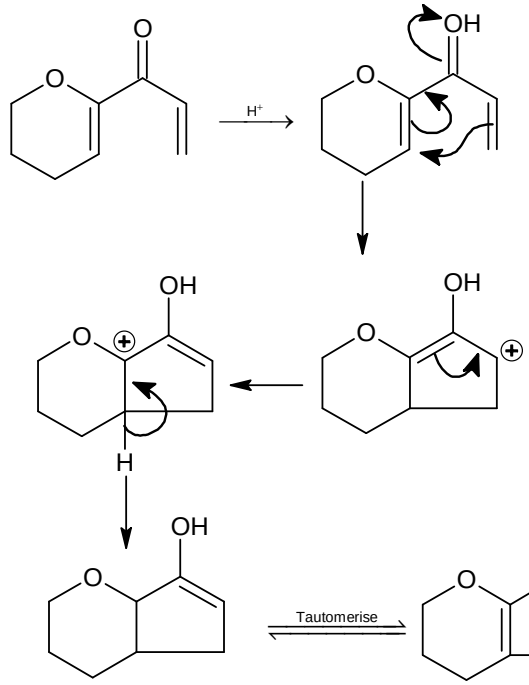
So, $V_x = V_y = 2 \Rightarrow V = 2\sqrt{2}$

$\therefore R_C = \frac{V^2}{a_N} = \frac{4\times 2}{4.\cos 45^o} = 2\sqrt{2}$ m = $\frac{4}{\sqrt{2}}$m

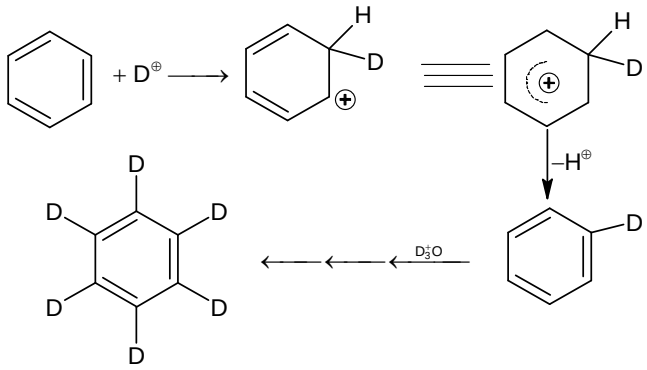
# Chemistry

PART – II

2.

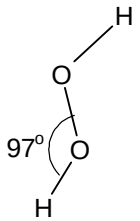



3.




4. Talc has empirical formula $Mg_2(Si_2O_5)_2Mg(OH)_2$ which $\left(Si_2O_5\right)_n^{2n-}$ units of two dimensional sheet silicates.

5.




It has half open book like structure having dihedral angle $97^o$

6. $\Delta_r H^o = \left(\left(2\times 46.1\right)+187.8-\left(2\times 285.8\right)+\Delta_r H^o\left(N_2H_4\right)\right)kJ/mol^{-1}$

= - 241.0 kJ/mol$^{-1}$

$\Delta H_{r^o}\left(N_2H_4\right) = 50.6\ kJ/mol$

Decomposition enthalpy = - 50.6 kJ/mol.

9. Radius ratio for MgO is = 0.464 which reflect the coordination number 6 and radius ratio for MgS is 0.353. Which reflect the coordination number 4.

10. $H_2O(\ell) + ClO_3^-(aq) \longrightarrow ClO_4^-(aq) + 2H^+(aq) + 2e^-$

$2H^+(aq) + ClO_3^-(aq) + 2e^- \longrightarrow ClO_2^-(aq) + H_2O(\ell)$

$2ClO_3^-(aq) \longrightarrow ClO_2^-(aq) + ClO_4^-(aq)$

$E^o_{cell} = 0.33 - 0.36 = -0.03$

$E^o_{cell} = \frac{RT}{nF}\ln K$ (at equilibrium)

$-0.03 = \frac{0.059}{2}\log K$

logK = - 1

K = 0.1

| | $2ClO_3^- \rightleftharpoons$ | $ClO_4^-$ | + $ClO_2^-$ |
|---|---|---|---|
| Initially | .1 | 0 | 0 |
| equilibrium | .1 – 2x | x | x |

$K = \frac{x^2}{(.1-2x)^2}$

$\frac{1}{10} = \frac{x^2}{(.1-2x)^2} \Rightarrow x = 0.019$ M

12. Probability of finding an electron is zero, i.e. $\psi^2 = 0$.

$$\left[\frac{1}{16\sqrt{4}}\left(\frac{1}{a_0}\right)^{3/2}\left[(\sigma-1)(\sigma^2-8\sigma+12)\right]e^{-\sigma/2}\right]^2 = 0$$

$(\sigma-1)(\sigma^2-8\sigma-12) = 0$

On solving for $\sigma$ we get

$\sigma$ = 1, 6, 2 and $\sigma = \frac{2r}{a_0}$

For $\sigma$ = 1; $1 = \frac{2r}{a_0}$ $r \Rightarrow \frac{a_0}{2}$

$\sigma = 2$; $2 = \frac{2r}{a_0}$ $r \Rightarrow a_0$

$\sigma = 6$; $6 = \frac{2r}{a_0}$ $r = 3a_0$

13. n-factor of $H_3PO_2$ is 4/3 for the reaction.

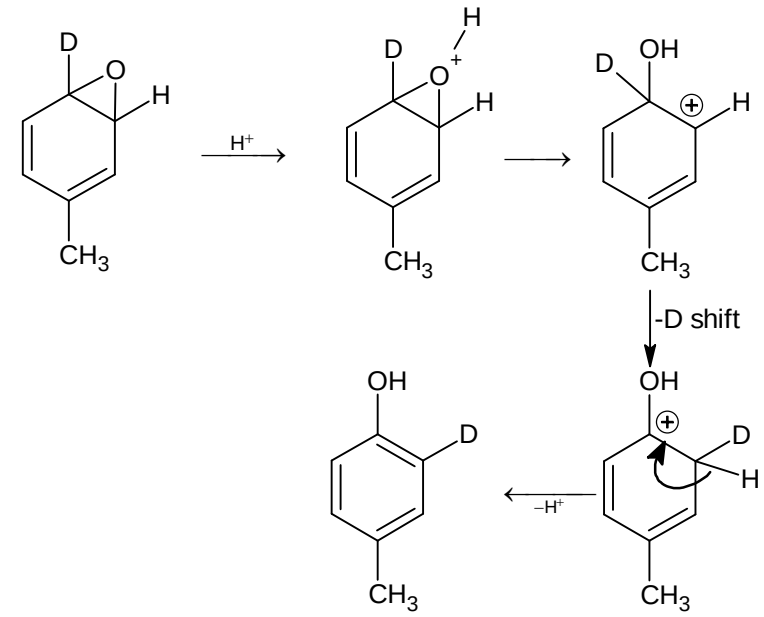
14. $\Delta T_b = K_b \times m \times i$

2.08 = 0.52 × 1 × i

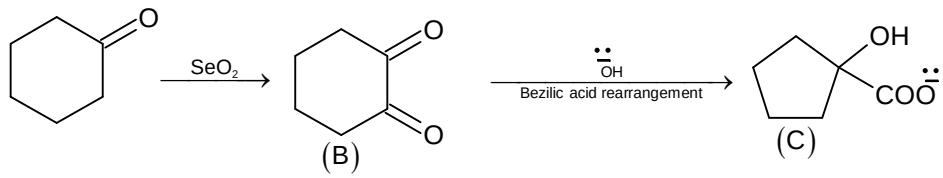
i = 4

This implies that the salt on dissociation gives 4 ions. Thus, the salt that gives four ions is $K_3[Fe(CN)_6]$.

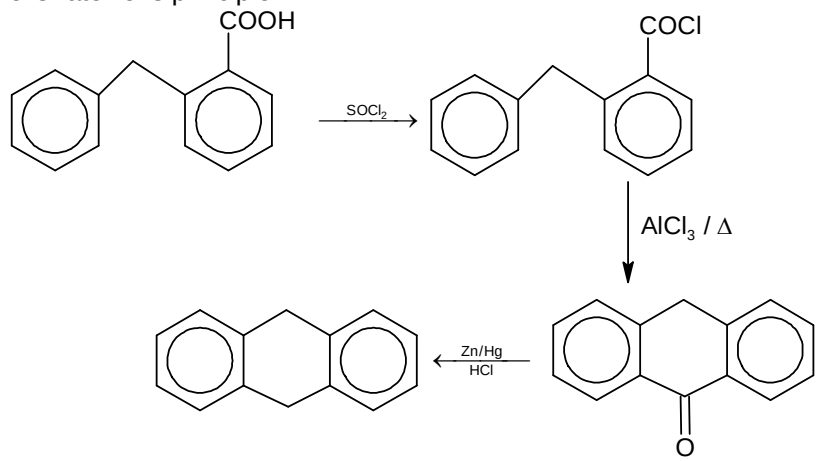
15.




16.




18. It is a non-degradable pollutant.
19. Colloidal silver bromide solution is used in photography.
20. Stability of carbonates increases down the group. Carbonates of alkali metals are more stable than those of alkali metals.
21. $\pi_{2px}$ and $\pi_{2py}$ have one nodal plane each. In $\sigma$1s there is no nodal plane and in case of $\pi^*_{2px}$ there are two nodal planes.
22. Le-Chatellier's principle.
25.




26. $NH_4Cl + \text{Nesseler's reagent} \longrightarrow \text{Brown colour of iodide million's base.}$

# Mathematics

PART – III

1. Using $T = S_1$ take mid–point (h, k) then equation of PQ is
$xh + yk – (h^2 + k^2) = 0$ ..... (1)
$mx – y + 2m = 0$ ..... (2)
Compare (1) and (2) eliminate m

2. As f(x) is continuous decreasing function on $(0, \infty)$, so $0 < f(8x) < f(6x) < f(4x)$
So $\lim\limits_{x\to\infty} 1 < \lim\limits_{x\to\infty} \frac{f(6x)}{f(8x)} < \lim\limits_{x\to\infty} \frac{f(4x)}{f(8x)}$
By squeeze play theorem $\lim\limits_{x\to\infty} \frac{f(6x)}{f(8x)} = 1$

3. The given lines will be parallel to lines $ax^2 – 6xy + y^2 = 0$
So, $\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + a = 0$
$m + m^2 = 6$ ..... (1)
Which gives m = –3 or 2
and $mm^2 = a$ ..... (2)
From (2) a = –27 or 8
Hence sum of all possible value of a = –19

4. $e^{\lim\limits_{x\to\infty} \frac{4}{x^2}\left[\ln g(2-x^2) - \ln g(2)\right]}$ $\left(\frac{0}{0} \text{ form}\right)$
$e^{\lim\limits_{x\to 0} \frac{\frac{g'(2-x^2)}{g(2-x^2)}\cdot(-2x)}{2x}} = e^{-\frac{g'(2)}{g(2)}} = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$

5. $I = \lim\limits_{n\to\infty} \frac{\int_0^{1/n} f(t)\,dt}{\frac{1}{n^2}}$ $\left(\frac{0}{0} \text{ form}\right)$
Using : L' Hospital and using Leibnitz rule
$I = \lim\limits_{n\to\infty} \frac{n^3 f\left(\frac{1}{n}\right)\left(-\frac{1}{n^2}\right)}{-2} = \lim\limits_{n\to\infty} \frac{n}{2} f\left(\frac{1}{n}\right)$
Put $n = \frac{1}{h}$ hence
$I = \frac{1}{2} \times \lim\limits_{h\to 0}\left(\frac{f(0+h) - f(0)}{h}\right) = \frac{1}{2} f'(0)$

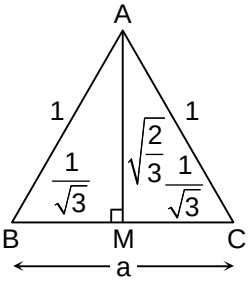
6. For $xy = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$
Slope of normal to xy = 1 for $x^2 > 0$ $(x \neq 0)$
Slope of line (given) = $\frac{\log_2(1+5a-a^2)}{5} > 0$
So $1 + 5a – a^2 > 1$

$a^2 - 5a < 0$ hence $a \in (0, 5)$

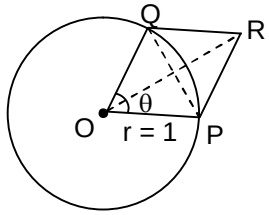
7. $a = \dfrac{2}{\sqrt{3}}$

$\Delta = \dfrac{1}{2}ah = \dfrac{1}{2} \times \dfrac{2}{\sqrt{3}} \times \dfrac{\sqrt{2}}{\sqrt{3}} = \dfrac{\sqrt{2}}{3}$

$R = \dfrac{abc}{4\Delta} = \dfrac{2(1)(1)(3)}{\sqrt{3}(4)\sqrt{2}} = \dfrac{\sqrt{3}}{2\sqrt{2}}$




8. Area of the parallelogram outside the circle
= Area of parallelogram – Area of sector OPQ
$= \left(\dfrac{1}{2}\sin\theta\right)2 - \dfrac{1}{2}\theta = \sin\theta - \dfrac{\theta}{2} = f(\theta)$




9. Using ${}^{n}C_r = \dfrac{n}{r} \cdot {}^{n-1}C_{r-1} = \dfrac{n(n-1)}{r(r-1)} \times {}^{n-2}C_{r-2}$ is equal to

10. We have $|(A^{-1})\ \text{adj}(B^{-1})\ \text{adj}(2A^{-1})| = \dfrac{1}{|A|} \cdot \dfrac{1}{|B|^2} \cdot \dfrac{64}{|A|^2} = \dfrac{64}{-2 \times 1 \times 4} = -8$

(By using $|A^{-1}| = \dfrac{1}{|A|}$ and $|\text{adj } B| = |B|^{n-1}$ where n is order of square matrix)

11. This distance between the given parallel lines (h) is $\dfrac{1}{\sqrt{5}}$

length of the side of triangle is = $\dfrac{2h}{\sqrt{3}}$

Area of triangle = $\dfrac{\sqrt{3}}{4} \times \dfrac{4h^2}{3} = \dfrac{h^2}{\sqrt{3}} = \dfrac{1}{5\sqrt{3}}$

12. The volume of tetrahedron = $\dfrac{1}{6}\left|\left[\overrightarrow{OA}\ \ \overrightarrow{OB}\ \ \overrightarrow{OC}\right]\right| = \dfrac{1}{6}\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \dfrac{1}{6}$

Area of base = $\dfrac{1}{2}\left|(\hat{i} + \hat{j} - \hat{i}) \times (\hat{j} + \hat{k} - \hat{i})\right| = \dfrac{1}{2}|\hat{i} \times \hat{k}| = \dfrac{1}{2}(\sqrt{2}) = \dfrac{1}{\sqrt{2}}$

Hence height = $\dfrac{3 \times \text{volume}}{\text{Area of base}} = \dfrac{3\sqrt{2}}{6} = \dfrac{1}{\sqrt{2}}$

13. $C = 20$

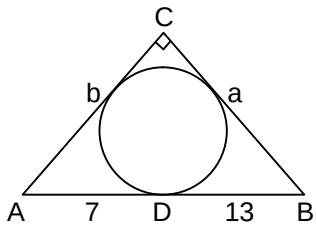
$\Delta = \dfrac{1}{2}ab$

$S - a = 7$

$S - b = 13$

$a - b = 6$ ..... (1)

$a^2 + b^2 = 400$



$(a-b)^2 + 2ab = 400$

$\Rightarrow ab = 182$

$\Delta = \frac{1}{2}ab = 91$

14. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Comparing $x = t^2 + t - 2 \Rightarrow y = t + 2 \Rightarrow z = t^2 + 6t + 9$

Hence $z - x = 5t + 11 \Rightarrow z - x = 5(y-2) + 11 \Rightarrow x + 5y - z + 1 = 0$

16. $\left[\vec{a}\ \ \vec{b}\ \ \vec{a}\times\vec{b}\right] = 8$

$\Rightarrow |\vec{a}\times\vec{b}|^2 = 8$

$|\vec{a}|^2 |\vec{b}|^2 \sin^2\theta = 8$

$|b|^2 = \frac{8}{1^2 \times \frac{1}{2}} = 16$

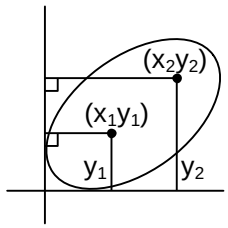
$|\vec{b}| = 4$

17. If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$

So $x_1x_2 = y_1y_2 = b^2$

$3d = 49$

$\sqrt{3d} = 7$




18. $f'(x) = \sec x - \operatorname{cosec} x$

$f'(x) = 0$ at $x = \frac{\pi}{4}$

Also $f''\left(\frac{\pi}{4}\right) > 0$ so f(x) is minimum at $x = \frac{\pi}{4}$

$f\left(\frac{\pi}{4}\right) = \int_0^{\pi/4} \sec\theta\, d\theta + \int_{\pi/4}^{\pi/2} \operatorname{cosec}\theta\, d\theta = \ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) = 2\ln(\sqrt{2}+1)$

19. $\int_0^a \ln(\cot a + \tan x)\,dx = \int_0^a \ln\left(\frac{\cos(a-x)}{\sin a \cdot \cos x}\right)dx$

$= \int_0^a \ln\cos(a-x)\,dx - \int_0^a \ln\sin a\,dx - \int_0^a \ln\cos x\,dx$

$I(a) = -\ln(\sin a)\cdot a$

$I(1) = -\ln(\sin a) = \ln|\operatorname{cosec} 1|$

20. Let $A = \lim_{x\to 0}\left(2^{x-1} + \frac{1}{2}\right)^{1/x}$ ($1^\infty$ form)

$= e^{\lim_{x\to 0}\frac{1}{x}\left[2^{x-1}+\frac{1}{2}-1\right]} = e^{\lim_{x\to 0}\left(\frac{2^x-1}{2x}\right)} = e^{\frac{1}{2}\ln 2} = \sqrt{2}$

21. $U_n = \dfrac{1}{(n+1)(n+2)} = \dfrac{(n+2)-(n+1)}{(n+1)(n+2)} = \dfrac{1}{n+1} - \dfrac{1}{n+2}$

$S_n = \sum_{n=1}^{n}\left(\dfrac{1}{n+1} - \dfrac{1}{n+2}\right) = \dfrac{1}{2} - \dfrac{1}{n+2}$

Hence $\lim\limits_{n\to\infty} S_n = \dfrac{1}{2}$

22. $z = \left(1+i\sqrt{3}\right)^{100}$

$= 2^{100}\left[\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right]^{100}$

$= 2^{100}\left[\cos\dfrac{100\pi}{3} + i\sin\dfrac{100\pi}{3}\right]$

$= 2^{100}\left[-\cos\dfrac{\pi}{3} - i\sin\dfrac{\pi}{3}\right]$

$= 2^{100}\left[-\dfrac{1}{2} - i\dfrac{\sqrt{3}}{2}\right]$

$\dfrac{\text{Re}(z)}{\text{Im}(z)} = \dfrac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \dfrac{1}{\sqrt{3}}$

23. $|wz - 1 - w^2| = a$
$|z + 1| = a$
$\Rightarrow |a - 1 + 2| = a$
$|z - 1| + 2 \geq a$
$0 \leq a \leq 4$

24. Since RHS is even. So LHS is also even integer so either both x and y should even or both should odd integer. Now
$x^4 - y^4 = (x - y)(x + y)(x^2 + y^2)$
$(x - y)$, $(x + y)$, $(x^2 + y^2)$ must be an even integer
Therefore $(x - y)(x + y)(x^2 + y^2)$ must be divisible by 8
But RHS is not divisible by 8
So given equation has no solution

25. Given numbers can be rearranged as
1, 4, 7 ….. $3n - 2 \rightarrow (3\lambda - 2)$ type
2, 5, 8 ….. $3n - 1 \rightarrow (3\lambda - 1)$ type
3, 6, 9 ….. $3n \rightarrow 3\lambda$ type
This means either both the numbers should from ($3\lambda$ type) or one number each from ($3\lambda - 2$) type and ($3\lambda - 1$) type

So total number of ways ${}^nC_2 + {}^nC_1 \times {}^nC_1 = \dfrac{n(n-1)}{2} + n^2 = \dfrac{3n^2 - n}{2}$

26. $(1 + x + x^2 + x^3)^5 = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ..... + a_{15}x^{15}$

Put x = 1 and x = –1 and add

$4^5 = a_0 + a_1 + a_2 + a_3 + a_4 + ..... + a^{15}$

$0 = a_0 - a_1 + a_2 - a_3 + a_4 - ..... - a^{15}$

---

$4^5 = 2(a_0 + a_2 + a_4 + ..... + a_{14})$

$a_0 + a_2 + a_4 + ..... + a_{14} = 512$

27. $f'(x) = 3x^2 + 2ax + b$

$y = f(x)$ is increasing $\forall\ x \in R$

$f'(x) \geq 0\ \forall\ x \in R$

So $(2a)^2 - 4 \times 3 \times b \leq 0$

$a^2 - 3b \leq 0$

This is true for exactly 16 ordered pairs (a, b) $1 \leq a, b \leq 6$

So required probability = $\frac{16}{36} = \frac{4}{9}$

28. $\frac{ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2}{6} \geq (a^6b^6c^6)^{1/6}$

So $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) \geq 6abc$

29. $\sum \tan^{-1}\left(\frac{2^r - 2^{r-1}}{1 + 2^r \cdot 2^{r-1}}\right)$

$= \sum_{r=1}^{n} \left[\tan^{-1}(2^r) - \tan^{-1}(2^{r-1})\right]$

$= \tan^{-1}(2^n) - \tan^{-1}(1)$

$= \tan^{-1}(2^n) - \frac{\pi}{4}$

30. $\frac{\cot A + \cot C}{\cot B} = \frac{\sin(A + C) \cdot \sin B}{\sin A \cdot \sin C \cdot \cos B}$

$= \frac{\sin^2 B}{\sin A \cdot \sin C \cdot \cos B}$

$= \frac{4R^2b^2}{4R^2ac\cos B}$

$= \frac{b^2}{ac\cos B}$

$= \frac{2b^2}{2002b^2 - b^2} = \frac{2}{2001}$